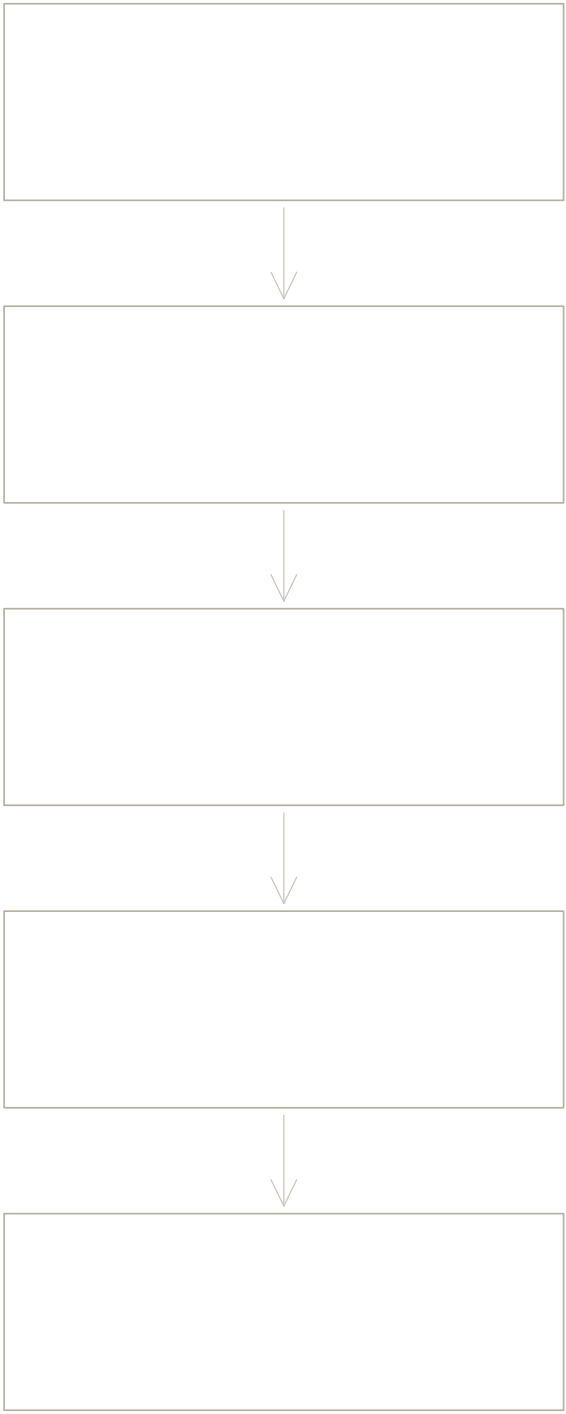Five-step vertical flow